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Antibiotics

Glycopeptides as vancomycin

inh DNA syn → Quinolones
inh RNA syn → Rifampin

Cross resistance
① organisim newly become resistant
② organisim resistant to one type of antibiotic and become resistant to another type even if not used against it due to structure similarity

Bacteria transfers resistance genes to other by :-
1) plasmid → conjugation
2) bacteriophage → Transduction
3) Transformation

* الأساسي اني اعمل culture بس ساعات مش ممكن اعمل empirical على اساس العمر , hepatic fnc , renal func , "

AntiBiotics

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* Uses of antibiotics :- (V. ImP)
① Targeted ② Emperical ③ Proplylactic

* Penicillins cross BBB in inflammations (meningitis)

* Antagonistic effect as :-
Bacteriostatic + Bactericidal
ex: Tetracyclines + penicillin
-aspenicillin need growing organism to work

* penicillin V is the only oral natural penicillin

* Anti-pseudomonal
. piperacillin or piperacillin / Tazobactam

. Cefepime
. Monobactam "Aztreonam"
. Ceftolozane / Tazobactam
. Polymyxins ستل

* Clarithromycin can be used in H-pylori

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* Antibiotic misuse can lead to
1) Resistance
2) superinfection

* Anti-metabolite → Trimethoprim, Sulfonalamide

* Procaine benzylpenicillin
Benzathine penicillin → IM

* 3rd, 4th, 5th generations are the most important due to their spectrum

* Any Antibiotic class اعرف :-
1) MOA
2) spectrum
3) SE ~~bold~~ موجودين في جدول و معمول
الحاجات المشهورة الي بتيجي
4) Uses xx

* Carbapenems resistant to pencillinase.

* Glycopeptides اخر مجموعة تشتغل على
ال cell wall

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Used in resistant enterococci and MRSA.
bacteriostatic with MRSA

* Oral bioavailability of vancomycin is very low so used for GIT infections as Clostridium difficle (pseudomembranous colitis)

* Clarithromycin can be used in H-pylori

* Lincosamides inhibit protein synthesis by binding to proteins

* Clindamycin : Antibacterial + Antitoxin used in Septecima and toxemia.
Toxic shock syndrome type of septicemia

* Aminoglycosides can't pentrate BBB only way to ttt meningitis → intrathecal

* Streptomycin used for TB and Brucella

* Fluroquinolones good oral bioavailability, good tissue pentration, wide spectrum

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Most common Quinolones

Second | Third | Fourth

2 | 1 | 3

4 | | ووادرتاي

5 |

* Linezolid used in VRSA

* Ansamycins narrow spectrum effective against mycobacteria

عشان كدة بيحاولوا يحافظوا عليها لل
مبينزلوش السوق عشان ميحصلش منهم
resistance

Rifaximin الوحيد الى بينزل
due to its poor oral bioavailability it's used as local anti-microbial as vancomycin. used ,

In Hepatic encephalopathy
we give Rifaximin not vancomycin
زمان كنا بنعي neomycin بس بطلنا
لانه لو استخدم لفترة طويلة يحصل
nephrotoxicity, ototoxicity

. They work on RNA polymerase

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* Metronidazole used in H-pylori

* Polymyxin effective against MDR as:-
Acinetobacter, Pseudomonas, klebsiella

المشكلة ان المصريين عملوا منه دوا شرب
لعلاج الإسهال في الأطفال بيطلع OTC

* جدول الـ SE اعرف في كل class
الـ bold

* Tetracycline not used in pregnant woman.
* Quinolones thought to cause tendinosis
so not given in children <18 y.o
قالوا بقالنا فترة مشوفناش العالدت دي
فممكن نجرب الـ ciprofloxacin مع الأطفال

* Vancomycin can cause Red man syndrome
Not a hypersensitivity reaction don't
give corticosone, epinephrine but Decrease
rate of vancomycin intake

* Chloramphenicol → bone marrow Aplasia

Categories of drugs in pregnancy

A	B	C	D
Safest	Safe	safe but use when necessary	not safe Avoided

* Nitrofurantoin not given in last months of pregnancy

* Sulfonamide can't be given as pregnant woman must take folate during pregnancy so Contraindicated in first 3Mo.
In last 3Mo inc risk of cholestasis in both mother, child
ممكن اديه في التلت شهور الي في النص
بس طالما في حاجات safe اصلا "مش بتاعه

* Streptogramin
Non-venous SE → Conjugated hyperbillirubinemia

* اركز في ال use في الأدوية الجديدة *

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plazomycin activity ال مقالش

CTAT :- complicated intra abdominal infection

* Ceftazidime / Avibactam
in Egypt called (Zavicefta)
Egypt used it in Pneumonia